Quantum Computing – Easy Study Answers (NRIA23)

PART A Answers

Qubit: A qubit is the basic unit of quantum information. It can be in a state $\alpha|0$■ + $\beta|1$■ where $\alpha^2 + \beta^2 = 1$.

Classical vs Quantum Gates: Classical gates are irreversible (like AND), quantum gates are reversible and use superposition.

Hilbert Space: A mathematical space where quantum states exist.

Born Rule: Probability = $|\text{amplitude}|^2$.

Hadamard Gate: Creates superposition.

Bell States: Special entangled two-qubit states.

Deutsch–Jozsa: Determines if function is constant or balanced.

Grover: $O(\sqrt{N})$ complexity.

Quantum Error Correction: Protects quantum info from errors.

BB84: Used for secure communication.

Bits vs Qubits

A classical bit is either 0 or 1. A qubit can be both at the same time due to superposition. This allows quantum computers to process many possibilities at once. Quantum gates manipulate qubits using reversible operations.

Math & Physics Contribution

Mathematics provides linear algebra concepts like vectors and matrices. Physics explains quantum behavior like superposition and entanglement. Together they form the foundation of quantum computing.

Hilbert Space

A Hilbert space is a vector space used to describe quantum states. It supports operations like inner product and allows representation of qubits mathematically.

Measurement Probability

When we measure a qubit $\alpha|0$■ + $\beta|1$■, probability of 0 is $|\alpha|^2$ and 1 is $|\beta|^2$. After measurement, state collapses.

Bloch Sphere

It is a 3D representation of a qubit. Any point on the sphere represents a quantum state.

Qubit Implementations

Qubits can be implemented using superconducting circuits, trapped ions, or photons. Each has advantages like stability or scalability.

Grover vs Classical

Classical search takes $O(N)$. Grover takes $O(\sqrt{N})$, making it faster for large datasets.

Complexity Classes

Classical uses P and NP. Quantum uses BQP which solves some problems faster than classical.

Quantum Error Correction

It protects quantum data using redundancy and special codes without directly copying qubits.

Quantum vs Classical Info

Classical info uses bits. Quantum uses qubits with superposition and entanglement, enabling more powerful computation.